LP2(P2- A* GRID PATH)

Implement A star Algorithm for any game search problem

```
import heapq


def heuristic(x1, y1, x2, y2):
    return abs(x1 - x2) + abs(y1 - y2)


def astar(grid, start, goal):
    n, m = len(grid), len(grid[0])
    pq = []
    heapq.heappush(pq, (0, 0, start[0], start[1]))  # f, g, x, y
    visited = set()

    while pq:
        f, g, x, y = heapq.heappop(pq)

        if (x, y) == goal:
            print("\nReached Goal! Minimum cost =", g)
            return

        if (x, y) in visited:
            continue
        visited.add((x, y))
        for dx, dy in [(1,0), (-1,0), (0,1), (0,-1)]:
            nx, ny = x + dx, y + dy
            if 0 <= nx < n and 0 <= ny < m and grid[nx][ny] != '#':
                new_g = g + 1
```

```python
            new_f = new_g + heuristic(nx, ny, goal[0], goal[1])
            heapq.heappush(pq, (new_f, new_g, nx, ny))

# ---------------- MAIN PROGRAM ----------------

n = int(input("Enter rows: "))
m = int(input("Enter cols: "))
print("Enter grid row by row (use S, G, ., #):")
grid = []
start = goal = None

for i in range(n):
   row = input().split()
   grid.append(row)

   for j in range(m):
      if row[j] == 'S':
         start = (i, j)
      if row[j] == 'G':
         goal = (i, j)

astar(grid, start, goal)
```

☆ **A* Algorithm – Viva Questions**

1. What is A* algorithm?

A* is an informed search algorithm that finds the shortest path using **g(n) + h(n)**.

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2. What is the formula of A*?

$f(n) = g(n) + h(n)$

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3. What is g(n)?

Actual cost from start node to current node.

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4. What is h(n)?

Estimated cost from current node to goal.

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5. Why is A* called informed search?

Because it uses heuristic knowledge (h).

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6. What is heuristic function?

A function that estimates distance to goal.

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7. Which heuristic is used in grid problem?

Manhattan distance.

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8. Why Manhattan distance?

Because movement is allowed in 4 directions only.

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9. Is A* optimal?

Yes, if heuristic is admissible (never overestimates).

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10. Is A* complete?

Yes, it always finds a solution if one exists.

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11. What data structure is used in A*?

Priority queue (min heap).

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12. Difference between A* and BFS?

BFS	A*
No heuristic	Uses heuristic
Uninformed	Informed

BFS	A*
Slower	Faster

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13. Where is A* used?

- Games
- Google Maps
- Robotics
- AI navigation

□ Grid Path – Viva Questions

1. What is Grid Path problem?

It is a path-finding problem in a matrix from start to goal.

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2. What are elements in grid?

- S = Start
- G = Goal
- **= Obstacle**
- . = Free path

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3. What is the objective?

To find shortest path from S to G.

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4. What movements are allowed?

Up, Down, Left, Right.

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5. What is obstacle?

A blocked cell that cannot be crossed (#).

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6. Which algorithm is used?

BFS, Dijkstra, or A*.

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7. Why is A* used in grid?

Because it is faster and uses heuristic.

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8. What is heuristic in grid path?

Distance from current cell to goal.

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9. What is output of grid A*?

Shortest path cost or route from S to G.

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10. Real-life applications?

- Video games
- GPS navigation
- Robot movement

☆ **One-line Viva Answer**

A*:

"A* is a heuristic search algorithm that finds the shortest path using actual cost and estimated cost."

Grid Path:

"Grid path problem is finding the shortest path from start to goal in a matrix while avoiding obstacles."